

TOPCLIFFE, United Kingdom

WMO: 032650

Lat: 54.2047N

Lon: 1.3900W

Elev: 28

StdP: 100.99

Time Zone: 0.00 (EUW)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.9	(c) -3.4	(d) -6.5	(e) 2.2	(f) -3.0	(g) -5.1	(h) 2.5	(i) -1.6	(j) 14.8	(k) 8.8	(l) 13.3	(m) 7.7	(n) 1.9	(o) 130	(p) 0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.3	(c) 24.8	(d) 18.5	(e) 22.9	(f) 17.4	(g) 21.3	(h) 16.5	(i) 19.4	(j) 23.1	(k) 18.2	(l) 21.5	(m) 17.2	(n) 20.3	(o) 4.9	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	12.9	21.2	16.9	12.1	19.8	16.0	11.4	18.6	55.3	23.2	51.6	21.6	48.6	20.4	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.1	10.5	9.3	-8.7	28.5	2.8	1.9	-10.8	29.9	-12.4	31.0	-14.0	32.1	-16.0	33.5
			-8.9	21.9	2.8	1.7	-10.9	23.1	-12.5	24.1	-14.1	25.1	-16.2	26.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	3.9	4.3	5.8	8.0	11.1	13.8	15.7	15.5	13.3	10.1	6.3	4.0		
	DBStd	5.25	3.23	3.24	2.89	2.82	2.83	2.54	2.34	2.36	2.43	2.96	3.28	3.94		
	HDD10.0	921	189	161	134	72	21	2	0	0	2	34	117	188		
	HDD18.3	3291	448	393	389	309	223	138	87	92	151	255	362	444		
	CDD10.0	686	1	1	3	13	56	116	178	172	103	38	6	2		
	CDD18.3	16	0	0	0	0	0	2	7	6	1	0	0	0		
	CDH23.3	129	0	0	0	1	4	22	61	34	7	1	0	0		
	CDH26.7	14	0	0	0	0	0	2	7	4	0	0	0	0		
(14) Wind	WSAvg	4.3	4.7	4.7	4.7	4.3	4.5	4.0	3.8	4.0	4.0	4.3	4.4	4.5		
(15) Precipitation	PrecAvg	685	60	47	41	51	48	60	56	69	56	63	68	65		
	PrecMax	959	127	88	90	140	97	162	128	155	115	147	137	129		
	PrecMin	518	8	10	9	5	11	8	18	14	15	21	22	20		
	PrecStd	109	28	22	22	35	21	37	25	32	30	28	29	28		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	13.7	16.1	20.4	23.5	26.1	27.3	27.2	24.4	19.2	15.5	13.4		
		MCWB	11.0	11.3	11.3	13.9	17.9	19.9	18.5	20.1	18.2	16.1	13.5	11.7		
	2%	DB	11.5	12.2	14.0	17.1	21.0	23.0	25.1	23.7	21.5	17.2	13.8	12.2		
		MCWB	9.7	10.1	10.3	12.4	15.9	17.0	18.3	18.2	16.7	14.7	12.0	10.6		
	5%	DB	10.2	10.8	12.4	15.2	18.7	21.0	23.0	21.7	19.5	15.9	12.7	11.2		
		MCWB	8.4	8.8	9.5	11.1	14.1	16.1	17.4	17.0	15.5	13.7	10.9	9.7		
	10%	DB	9.1	9.6	11.1	13.5	16.8	19.3	21.1	20.4	18.0	14.9	11.5	10.0		
		MCWB	7.5	7.7	8.4	10.1	12.8	15.0	16.4	16.1	14.5	12.8	9.9	8.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	11.5	12.2	14.9	18.9	21.2	20.5	20.8	19.5	16.9	14.0	12.0		
		MCDB	12.5	13.1	14.7	18.2	21.2	23.9	25.4	25.8	23.2	18.8	15.3	13.0		
	2%	WB	9.9	10.4	10.9	13.1	16.6	18.2	19.1	19.0	17.5	15.3	12.4	11.0		
		MCDB	11.2	12.0	13.3	16.2	20.3	21.2	23.2	22.4	20.3	16.6	13.5	12.0		
	5%	WB	8.8	9.1	9.8	11.7	14.9	16.6	18.0	18.0	16.2	14.2	11.2	9.9		
		MCDB	10.0	10.6	12.0	14.5	18.1	19.9	21.7	20.8	18.4	15.6	12.4	11.1		
	10%	WB	7.6	8.0	8.7	10.5	13.4	15.5	16.9	16.9	15.2	13.2	10.1	8.6		
		MCDB	8.9	9.4	10.9	13.1	16.0	18.6	20.5	19.4	17.4	14.5	11.3	9.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	6.8	8.2	9.1	9.3	9.1	9.3	9.1	8.8	7.4	6.5	6.1		
		MCDBR	6.8	7.7	10.4	12.0	12.8	12.7	12.9	11.0	11.3	8.2	7.1	6.8		
	5% WB	MCWBR	6.1	6.3	7.3	7.8	8.3	7.7	6.8	6.2	7.1	6.1	6.0	6.0		
		MCWBR	6.9	7.3	9.6	11.0	12.0	11.2	11.3	9.6	9.9	7.8	6.7	6.5		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.337	0.368	0.396	0.395	0.398	0.399	0.386	0.366	0.348	0.322	0.307		
		taud	2.316	2.366	2.287	2.230	2.277	2.293	2.298	2.349	2.411	2.431	2.378	2.293		
	Ebn at Noon		624	751	801	828	851	850	841	830	801	729	623	555		
		Edh at Noon	58	76	102	123	124	123	120	108	90	71	54	49		
(44) All-Sky Solar Radiation	RadAvg		0.62	1.26	2.34	3.69	4.71	4.80	4.64	3.83	2.78	1.55	0.76	0.48		
	RadStd		0.07	0.14	0.30	0.41	0.27	0.42	0.42	0.25	0.25	0.13	0.08	0.05		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page